WEEK 3 POWER BI REPORT

Financial Data Import, Transformation & Time Intelligence

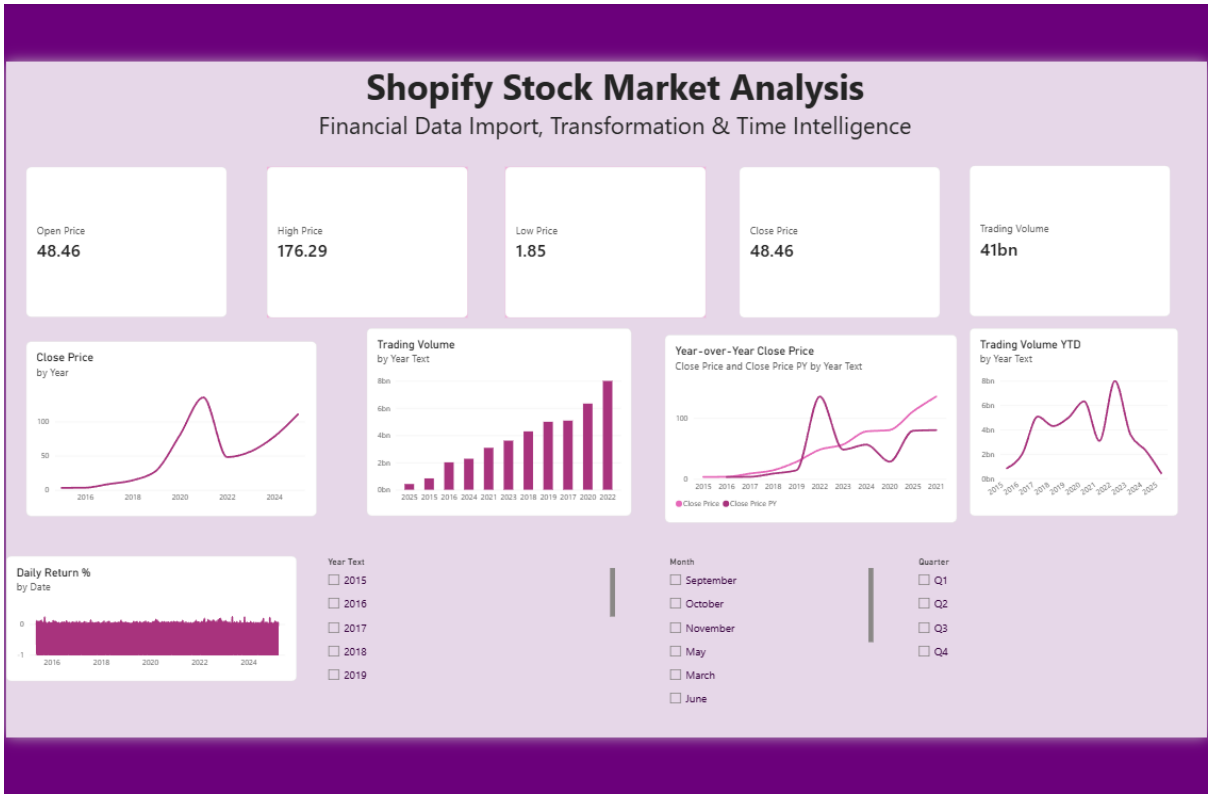
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Introduction:

This project focuses on importing, transforming, and analyzing financial data using Microsoft Power BI. The dataset contains stock market information such as opening price, highest price, lowest price, closing price, adjusted closing price, and trading volume.

A date table was created to perform time-based analysis. Different DAX measures were also created to calculate financial values and time-intelligence metrics such as previous closing price, daily return, year-over-year change, and year-to-date trading volume.

The final dashboard presents the financial information using interactive charts, cards, and slicers.



Objective

The main objectives of this project are:

- To import financial data into Power BI.
- To clean and transform the imported data.
- To create a separate Date Table for time-based analysis.
- To establish a relationship between the Date Table and financial data.
- To create DAX measures for financial calculations.
- To implement time-intelligence calculations.
- To design an interactive financial dashboard.
- To analyze stock prices and trading volume over different time periods.

Dataset Description

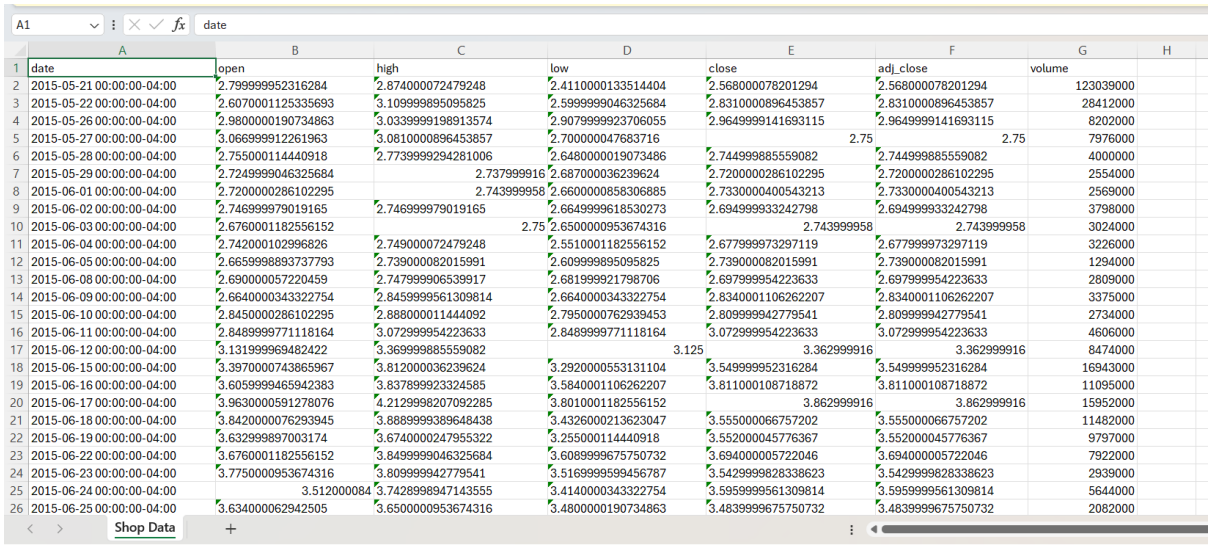
The financial dataset contains historical stock market information.

The main table used in the project is named Shop Data.

Important columns in the dataset:

Column	Description
Date	Date of the stock record
Open	Opening price of the stock
High	Highest price during the period
Low	Lowest price during the period
Close	Closing price of the stock
Adj Close	Adjusted closing price
Volume	Number of shares traded

The dataset contains daily financial records that can be analyzed across different years, quarters, and months.



	A	B	C	D	E	F	G	H
1	date	open	high	low	close	adj_close	volume	
2	2015-05-21 00:00:00-04:00	2.799999952316284	2.874000072479248	2.4110000133514404	2.568000078201294	2.568000078201294	123039000	
3	2015-05-22 00:00:00-04:00	2.6070001125335693	3.109999895095825	2.5999999046325684	2.8310000896453857	2.8310000896453857	28412000	
4	2015-05-26 00:00:00-04:00	2.9800000190734863	3.0339999198913574	2.9079999923706055	2.9649999141693115	2.9649999141693115	8202000	
5	2015-05-27 00:00:00-04:00	3.066999912261963	3.0810000896453857	2.700000047683716	2.75	2.75	7976000	
6	2015-05-28 00:00:00-04:00	2.755000114440918	2.7739999294281006	2.6480000019073486	2.744999885559082	2.744999885559082	4000000	
7	2015-05-29 00:00:00-04:00	2.7249999046325684	2.737999916	2.687000036239624	2.7200000286102295	2.7200000286102295	2554000	
8	2015-06-01 00:00:00-04:00	2.7200000286102295	2.743999958	2.6600000858306885	2.7330000400543213	2.7330000400543213	2569000	
9	2015-06-02 00:00:00-04:00	2.746999979019165	2.746999979019165	2.6649999618530273	2.694999933242798	2.694999933242798	3798000	
10	2015-06-03 00:00:00-04:00	2.6760001182556152	2.75	2.6500000953674316	2.743999958	2.743999958	3024000	
11	2015-06-04 00:00:00-04:00	2.742000102996826	2.749000072479248	2.5510001182556152	2.677999973297119	2.677999973297119	3226000	
12	2015-06-05 00:00:00-04:00	2.6659998893737793	2.739000082015991	2.609999895095825	2.739000082015991	2.739000082015991	1294000	
13	2015-06-08 00:00:00-04:00	2.690000057220459	2.747999906539917	2.681999921798706	2.697999954223633	2.697999954223633	2809000	
14	2015-06-09 00:00:00-04:00	2.6640000343322754	2.8459999561309814	2.6640000343322754	2.8340001106262207	2.8340001106262207	3375000	
15	2015-06-10 00:00:00-04:00	2.8450000286102295	2.888000011444092	2.7950000762939453	2.809999942779541	2.809999942779541	2734000	
16	2015-06-11 00:00:00-04:00	2.8489999771118164	3.072999954223633	2.8489999771118164	3.072999954223633	3.072999954223633	4606000	
17	2015-06-12 00:00:00-04:00	3.131999969482422	3.369999885559082	3.125	3.362999916	3.362999916	8474000	
18	2015-06-15 00:00:00-04:00	3.3970000743865967	3.812000036239624	3.2920000553131104	3.549999952316284	3.549999952316284	16943000	
19	2015-06-16 00:00:00-04:00	3.6059999465942383	3.837899923324585	3.5840001106262207	3.811000108718872	3.811000108718872	11095000	
20	2015-06-17 00:00:00-04:00	3.9630000591278076	4.2129998207092285	3.8010001182556152	3.862999916	3.862999916	15952000	
21	2015-06-18 00:00:00-04:00	3.8420000076293945	3.8889999389648438	3.4326000213623047	3.555000066757202	3.555000066757202	11482000	
22	2015-06-19 00:00:00-04:00	3.632999897003174	3.6740000247955322	3.255000114440918	3.552000045776367	3.552000045776367	9797000	
23	2015-06-22 00:00:00-04:00	3.6760001182556152	3.8499999046325684	3.6089999675750732	3.694000005722046	3.694000005722046	7922000	
24	2015-06-23 00:00:00-04:00	3.7750000953674316	3.809999942779541	3.5169999599456787	3.5429999828338623	3.5429999828338623	2939000	
25	2015-06-24 00:00:00-04:00	3.512000084	3.7428998947143555	3.4140000343322754	3.5959999561309814	3.5959999561309814	5644000	
26	2015-06-25 00:00:00-04:00	3.634000062942505	3.6500000953674316	3.4800000190734863	3.4839999675750732	3.4839999675750732	2082000	

Shop Data

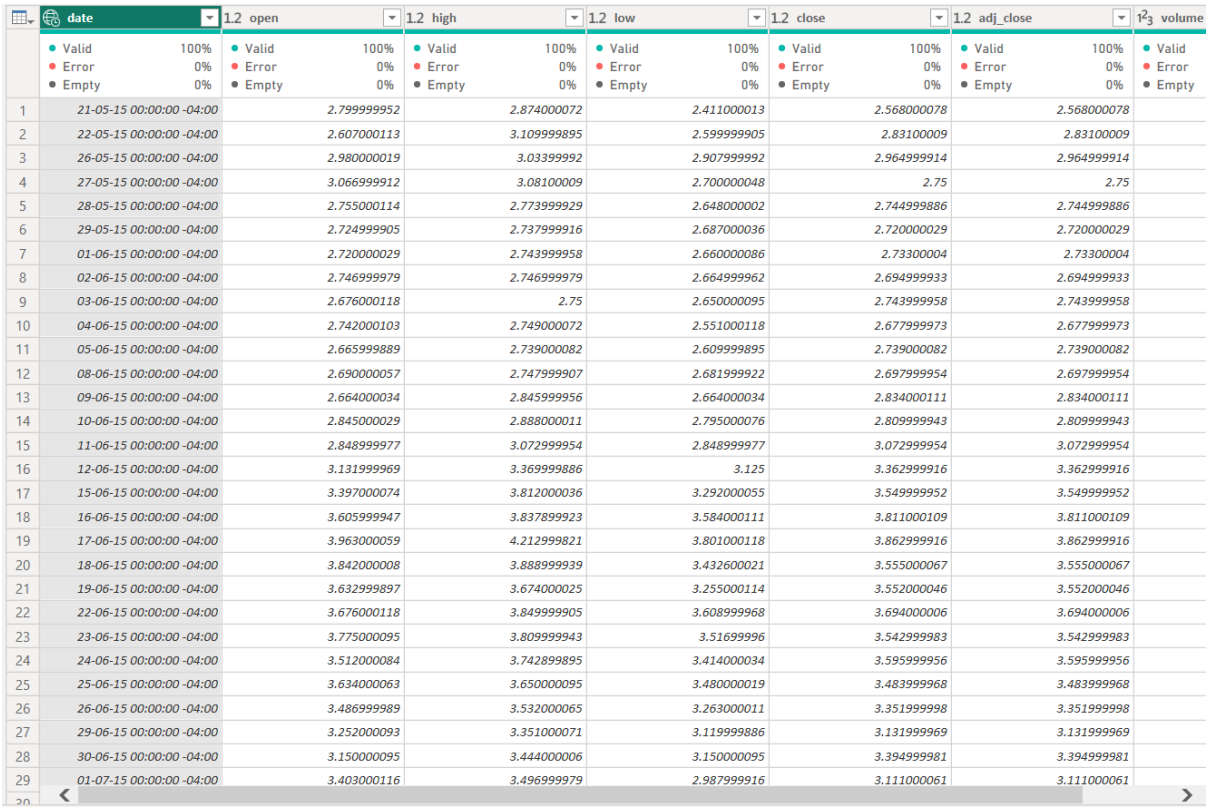
Data Import

The financial dataset was imported into Power BI Desktop.

Steps:

1. Open Power BI Desktop.
2. Select Get Data.
3. Select the appropriate file format.
4. Choose the financial dataset.
5. Load the data into Power BI.
6. Verify the columns and their data types.
7. Rename the main table as Shop Data for easier identification.

After importing the data, the table was checked to ensure that the financial columns were available correctly.



	date	open	high	low	close	adj_close	volume
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
1	21-05-15 00:00:00 -04:00	2.799999952	2.874000072	2.411000013	2.568000078	2.568000078	
2	22-05-15 00:00:00 -04:00	2.607000113	3.109999895	2.599999905	2.83100009	2.83100009	
3	26-05-15 00:00:00 -04:00	2.980000019	3.03399992	2.907999992	2.964999914	2.964999914	
4	27-05-15 00:00:00 -04:00	3.066999912	3.08100009	2.700000048	2.75	2.75	
5	28-05-15 00:00:00 -04:00	2.755000114	2.773999929	2.648000002	2.744999886	2.744999886	
6	29-05-15 00:00:00 -04:00	2.724999905	2.737999916	2.687000036	2.720000029	2.720000029	
7	01-06-15 00:00:00 -04:00	2.720000029	2.743999958	2.660000086	2.73300004	2.73300004	
8	02-06-15 00:00:00 -04:00	2.746999979	2.746999979	2.664999962	2.694999933	2.694999933	
9	03-06-15 00:00:00 -04:00	2.676000118	2.75	2.650000095	2.743999958	2.743999958	
10	04-06-15 00:00:00 -04:00	2.742000103	2.749000072	2.551000118	2.677999973	2.677999973	
11	05-06-15 00:00:00 -04:00	2.665999889	2.739000082	2.609999895	2.739000082	2.739000082	
12	08-06-15 00:00:00 -04:00	2.690000057	2.747999907	2.681999922	2.697999954	2.697999954	
13	09-06-15 00:00:00 -04:00	2.664000034	2.845999956	2.664000034	2.834000111	2.834000111	
14	10-06-15 00:00:00 -04:00	2.845000029	2.888000011	2.795000076	2.809999943	2.809999943	
15	11-06-15 00:00:00 -04:00	2.848999977	3.072999954	2.848999977	3.072999954	3.072999954	
16	12-06-15 00:00:00 -04:00	3.131999969	3.369999886	3.125	3.362999916	3.362999916	
17	15-06-15 00:00:00 -04:00	3.397000074	3.812000036	3.292000055	3.549999952	3.549999952	
18	16-06-15 00:00:00 -04:00	3.605999947	3.837899923	3.584000111	3.811000109	3.811000109	
19	17-06-15 00:00:00 -04:00	3.963000059	4.212999821	3.801000118	3.862999916	3.862999916	
20	18-06-15 00:00:00 -04:00	3.842000008	3.888999939	3.432600021	3.555000067	3.555000067	
21	19-06-15 00:00:00 -04:00	3.632999897	3.674000025	3.255000114	3.552000046	3.552000046	
22	22-06-15 00:00:00 -04:00	3.676000118	3.849999905	3.608999968	3.694000006	3.694000006	
23	23-06-15 00:00:00 -04:00	3.775000095	3.809999943	3.51699996	3.542999983	3.542999983	
24	24-06-15 00:00:00 -04:00	3.512000084	3.742899895	3.414000034	3.595999956	3.595999956	
25	25-06-15 00:00:00 -04:00	3.634000063	3.650000095	3.480000019	3.483999968	3.483999968	
26	26-06-15 00:00:00 -04:00	3.486999989	3.532000065	3.263000011	3.351999998	3.351999998	
27	29-06-15 00:00:00 -04:00	3.252000093	3.351000071	3.119999886	3.131999969	3.131999969	
28	30-06-15 00:00:00 -04:00	3.150000095	3.444000006	3.150000095	3.394999981	3.394999981	
29	01-07-15 00:00:00 -04:00	3.403000116	3.496999979	2.987999916	3.111000061	3.111000061	

Data Transformation

After importing the dataset, the data was reviewed and prepared for analysis.

The important financial columns were retained:

- Date
- Open
- High
- Low
- Close
- Adj Close
- Volume

The Date column was used as the basis for time-based analysis.

The main financial table was renamed to Shop Data to make the DAX formulas easier to understand and manage.

Creating the Date Table

A separate Date Table was created to perform time-intelligence calculations.

The following DAX formula was used:

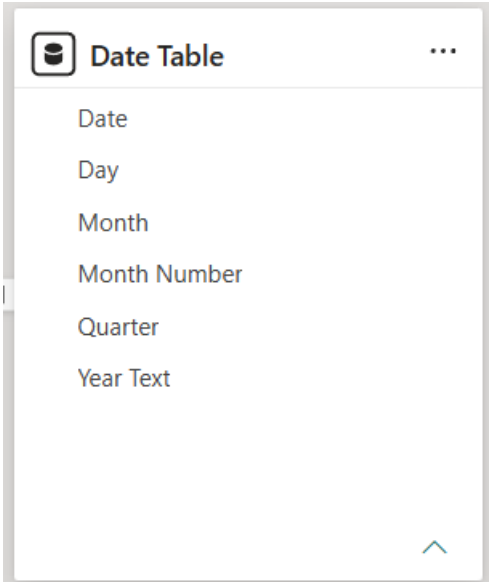
```
Date Table =
CALENDAR(
    MIN('Shop Data'[date]),
    MAX('Shop Data'[date])
)
```

This creates a continuous date table starting from the minimum date in the financial dataset and ending at the maximum date.

Additional columns were created:

```
Year Text = FORMAT('Date Table'[Date], "yyyy")
Quarter = "Q" & QUARTER('Date Table'[Date])
Month Number = MONTH('Date Table'[Date])
Month = FORMAT('Date Table'[Date], "MMMM")
Day = DAY('Date Table'[Date])
```

These columns allow the financial data to be analyzed by Year, Quarter, Month, and Day.



Creating the Relationship

A relationship was created between the Date Table and the financial table.

Relationship:

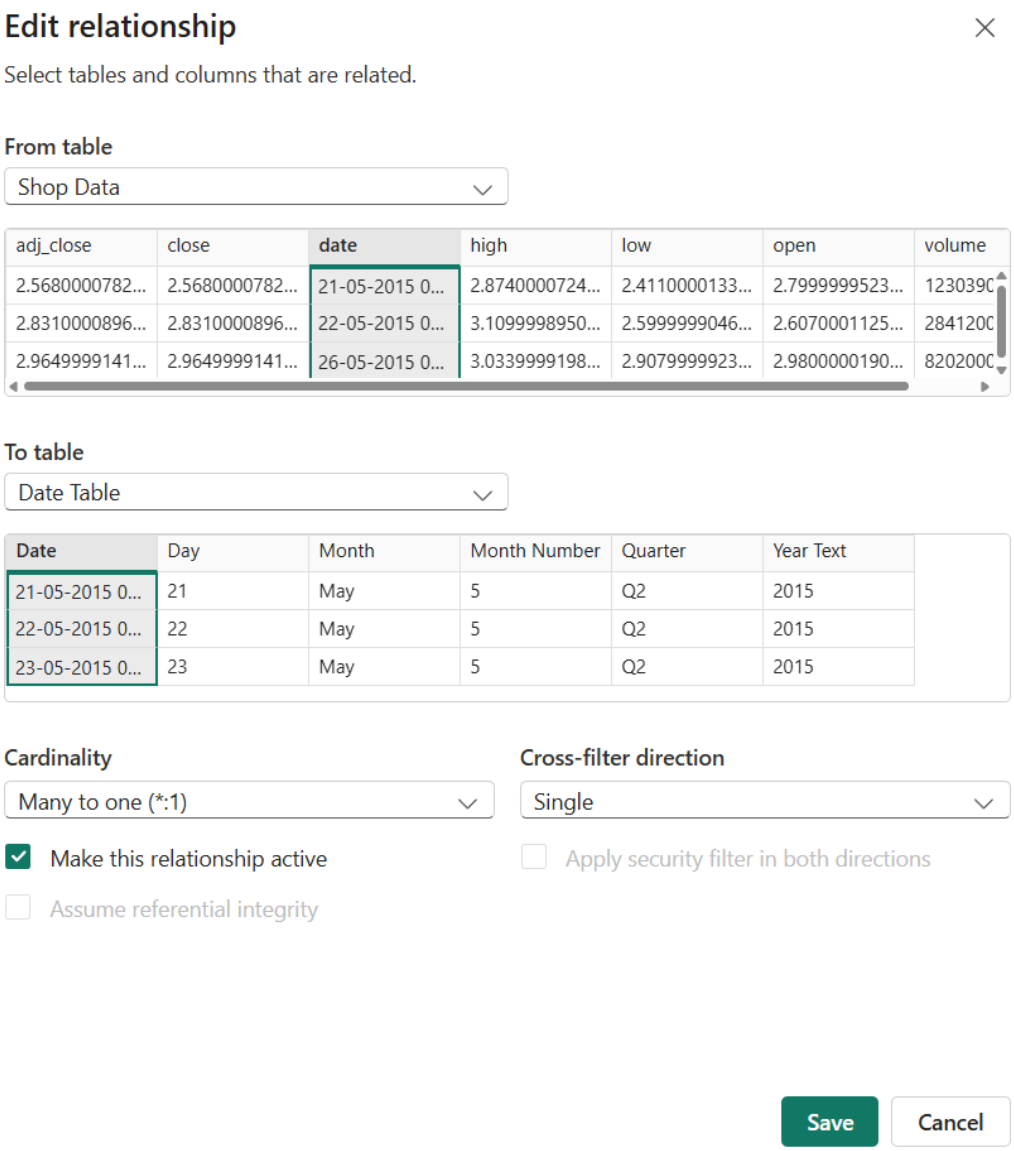
Date Table[Date] → Shop Data[date]

The relationship is:

Many-to-One (1:*)

The Date Table acts as the one side, while the Shop Data table contains multiple financial records corresponding to dates.

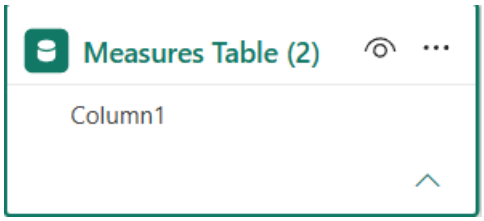
This relationship allows the Date Table to control the financial data during time-based analysis.



Creating the Measures Table

A separate Measures Table was used to organize the DAX measures.

Keeping the measures in a separate table makes the model easier to understand and keeps calculated measures organized.



Creating Financial Measures

The following DAX measures were created.

Open Price

Open Price = MAX('Shop Data'[open])

This measure calculates the maximum opening price available within the selected filter context.

High Price

High Price = MAX('Shop Data'[high])

This measure calculates the highest recorded price.

Low Price

Low Price = MIN('Shop Data'[low])

This measure calculates the lowest recorded price.

Close Price

Close Price = MAX('Shop Data'[close])

This measure calculates the closing price based on the selected filter context.

Trading Volume

Trading Volume = SUM('Shop Data'[volume])

This measure calculates the total trading volume.

Time Intelligence Measures

Time-intelligence calculations were created to compare financial values across different dates and years.

Previous Close

```
Previous Close =
CALCULATE(
    MAX('Shop Data'[close]),
    DATEADD('Date Table'[Date], -1, DAY)
)
```

This measure retrieves the closing price from the previous day.

Daily Return %

```
Daily Return % =
DIVIDE(
    MAX('Shop Data'[close]) - [Previous Close],
    [Previous Close],
    0
)
```

This measure calculates the percentage change between the current closing price and the previous closing price.

It helps visualize daily changes in the stock price.

Close Price PY

```
Close Price PY =
CALCULATE(
    [Close Price],
    SAMEPERIODLASTYEAR('Date Table'[Date])
)
```

This measure retrieves the closing price for the corresponding period of the previous year.

Close Price YoY%

```
Close Price YoY% =
DIVIDE(
```

[Close Price] - [Close Price PY],

[Close Price PY],

0

)

This measure calculates the percentage difference between the current closing price and the previous year's closing price.

Trading Volume YTD

Trading Volume YTD =

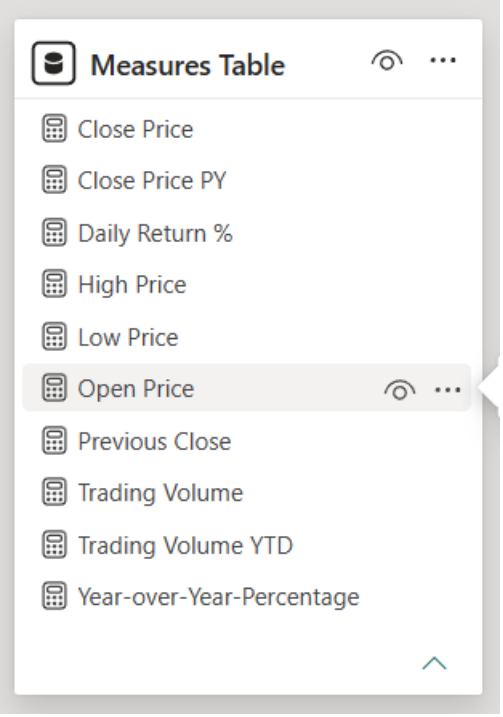
TOTALYTD(

[Trading Volume],

'Date Table'[Date]

)

This measure calculates the cumulative trading volume from the beginning of the year up to the selected date.



Creating the Dashboard

An interactive SHOPIFY STOCK MARKET ANALYSIS dashboard was created using KPI cards, charts, and slicers to present the financial data clearly.

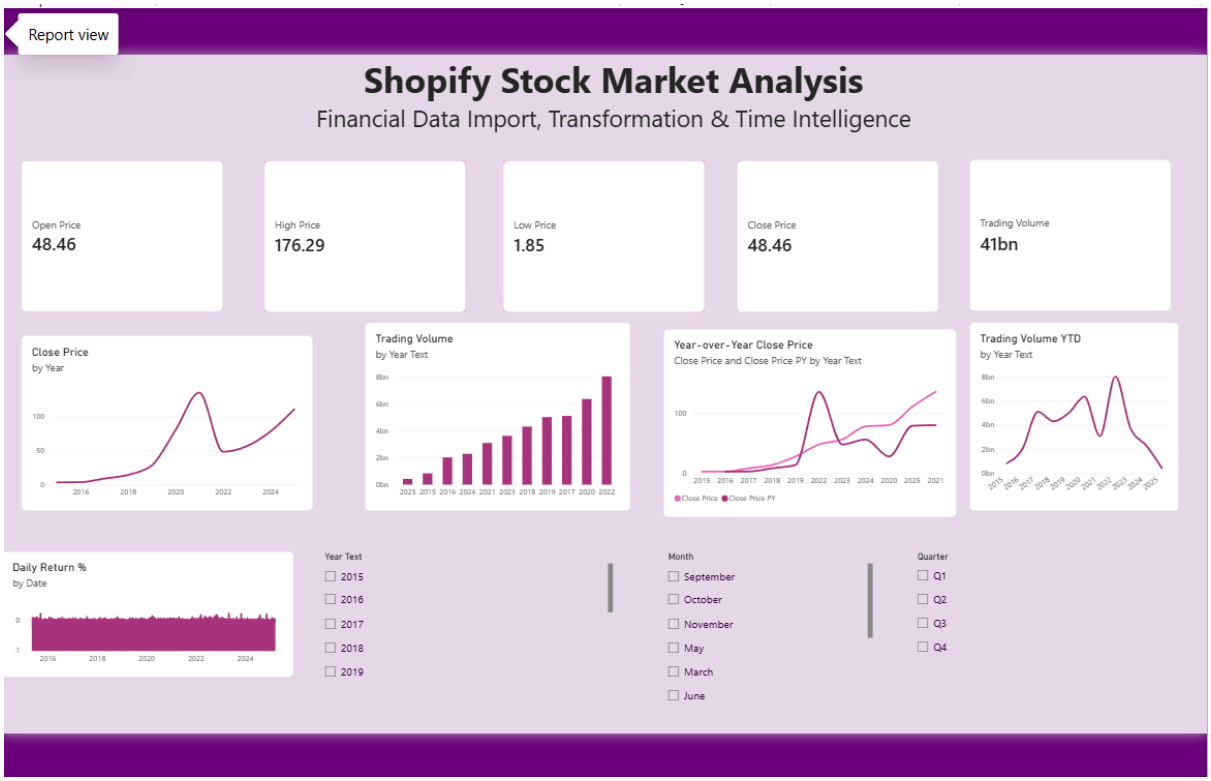
The dashboard includes:

- 5 KPI Cards**:** Open Price, High Price, Low Price, Close Price, and Trading Volume.
- Charts: Closing Price Trend, Year-over-Year Close Price, Trading Volume by Year, Trading Volume YTD, and Daily Return % Trend.
- Slicers: Year, Quarter, and Month for interactive filtering.

The dashboard was formatted with a light lavender background, dark purple header/footer, dark KPI cards, and contrasting chart elements for a clean appearance.

Dashboard Analysis

The completed dashboard combines financial measures and time-intelligence calculations to analyze stock prices and trading volume across different time periods. The slicers allow users to interactively filter the visualizations by year, quarter, and month, making it easier to observe trends and changes in the financial data.



Conclusion

This project demonstrated how Power BI can be used to import, transform, analyze, and visualize financial data.

A dedicated Date Table and relationship were created to support time-based analysis. DAX measures were developed for financial metrics and time intelligence, including previous-day closing price, daily return, year-over-year comparison, and year-to-date trading volume.

The final dashboard provides an interactive way to understand stock price movements and trading activity across different time periods.

Overall, the project helped in understanding data modeling, DAX calculations, time intelligence, dashboard design, and interactive data visualization in Power BI.